Credit Risk Modelling Proposal

Credit Risk Modelling is defined as the mathematical process of using previously saved data about a person that'll help us predict how likely it is that the person will pay the loan. In the context of Machine Learning, we use a concept called **"Default Risk"**, which helps us calculate the default probability that the person is less likely to pay their loan back, which is also called **"Defaulting"**. Even though this is mainly done by credit card companies, this is used in almost any loan, where the borrower has the ability to spend the money that can be used in the future.

In probability, we've a concept called the **"Law of Large Numbers"**. As we perform a large number of trials, the accuracy of predicted data increases. This is why large banks with millions of customers effectively utilize millions of data points, using ML models to find the smallest of patterns that small local lenders cannot find. By analyzing the possible outcomes, these models calculate the predicted risk score.

To conclude, "scaling" is very important in Machine Learning, especially in Credit Risk Systems where losses can be minimized and predictions can be done accurately.

Data Requirements

To help build a stable and an efficient Loan Management System, these are the variables that the system like this requires:

1. **Reliability**: The system should work accurately under normal conditions, and should identify failures and prevent crashing. Even if the ML model crashes, the remaining applications should still be accepted and be kept for review. This is called **"Graceful Degradation"**.
2. **Security**: Unauthorized access of data must be prevented, and the system must be protected from several fraudsters. This is a complete **"non-negotiable requirement"**, since all the bank details of a customer are extremely sensitive, hence these must be encrypted and tracked regularly.
3. **Integrity**: Making sure that the data used is free of errors.

4. **Scalability**: This means that the system must handle 10 users as it does for 10,000 users. You wouldn't need to rebuild the system or rewrite the code again.
5. **Consistency**: Multiple components of the system must come up with a unanimous conclusion when it comes to writing data. No two opinions can conflict and this is considered one of the biggest risks in financial systems.

Data Outputs

A loan officer would have to use these following important outputs to make an improved loan approval decision:

1. **Credit Score**: This represents the probability the borrower will stop paying. It lies in the probability between 0 and 1.
2. **Interest Rate**: Rate is connected directly to the credit score. If a system is producing a **"higher risk"**, it indicates a higher rate to make up for the required losses. It could also be due to certain factors such as maximizing the usage of credit cards.
3. **Detecting various fraud alerts**: This is extremely critical to avoid any legal and financial risks.
4. **Loan Eligibility**: Decisions to either approve or reject a loan with a valid reason. This is one of the most important factors.

Debt-to-Income Ratio (DTI): It's calculated as the ratio of monthly debt payments to the monthly income of an individual. Certain components that come under this could be rent, student loans etc.

Architecture

The various model types and architectures available for this machine learning system include:

1. **Logistic regression:** It's considered to be the simplest model. It just takes one look at your income and debts and outputs a 0-1 probability.
2. **Decision trees:** Think of it like a flowchart that contains a series of yes/no questions.
3. **Random Forest:** It's a combination of several decision trees. It's most accurate since it can identify missing data. This is used widely in banking.

4. **XGBoost / Gradient Boosting:** Trees that correct each other's mistakes. It's currently the **"industry's favorite"** for credit scoring. It works at a really high speed and provides the most structured accuracy for a large set of data.
5. **Neural Networks:** Models inspired by the brain. It finds hidden patterns humans wouldn't think to look for. This is particularly useful in large banks with multiple records.
6. **Naive Bayes Classifier:** It uses basic mathematical concepts of probability and is fast and simple. This is useful when data is limited.

XGBoost / Gradient Boosting will perform best. It handles the Law of Large Numbers perfectly and offers high prediction while seeing which data points influenced the result.

Risks and Challenges

While developing this credit risk modeling system, here are some of the risks and challenges we need to keep in mind:

1. **Model Bias**: If the data contains biases, then the model will learn those patterns and just copy them. The model doesn't know that there's a certain concept of **"fairness"**.
2. **Overfitting**: This happens when the model tends to learn a lot from the training data. But it doesn't tend to perform well on test data. This is much similar to a student who memorizes past practice material but is unable to handle a new exam format.
3. **Drift**: A model that was found to be 80% accurate in one year might only be 50% accurate in another year. This can happen due to several economic factors, hence performance metrics need to be evaluated on a regular basis.
4. **Quality of Data**: If past records have improper structures, incorrect data, and missing values, the model will end up learning wrong patterns and making bad predictions.
5. **Regulatory**: Lenders have to explain the reason for the rejection of their loan. Whatever decisions the model makes, it must be understandable to the regulators.